

Istanbul, Turkey

April 18, 2026

Professor Binshan Lin
Editor-in-Chief
Expert Systems with Applications

Dear Professor Lin,

I am pleased to submit our original research manuscript entitled “**Hybrid CP-SAT and Quantum-Inspired Optimization for Airline Disruption Recovery with Explainable Decision Support**” for consideration for publication in *Expert Systems with Applications*.

Airline disruption management is a critical operational challenge where exact mathematical optimization often struggles with real-time constraints, and metaheuristics struggle with strict regulatory feasibility. Our study presents three key innovations that address these gaps:

1. A novel hybrid architecture that couples Google OR-Tools CP-SAT with a Quantum-Inspired Genetic Algorithm (QIGA) via a warm-start mechanism, achieving a 5% profit increase over standalone exact methods.
2. A formal mathematical proof (Proposition 1) demonstrating the monotonic non-degradation property of the warm-start coupling, ensuring feasibility is maintained while refinement occurs.
3. A dual-layer explainability framework (SHAP + Constraint Reason Codes) that provides transparency aligned with the EASA AMC 20-42 regulatory guidelines.

We believe this work is well-aligned with the scope of *Expert Systems with Applications*, as it presents an intelligent system designed for complex, industrial combinatorial optimization and integrates modern explainability techniques for safety-critical decision support.

This manuscript is original work and has not been published elsewhere, nor is it currently under consideration by another journal. All authors have approved the manuscript for submission and have no conflicts of interest to disclose.

Thank you for your time and consideration of our work.

Sincerely,

Kürşat Kahya
Independent Researcher